Project: Worked examples
Title: Material strength margin
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the factors of safety of a material against an applied equivalent stress, on yield and, when given, on ultimate strength. It does not check fatigue or stability.

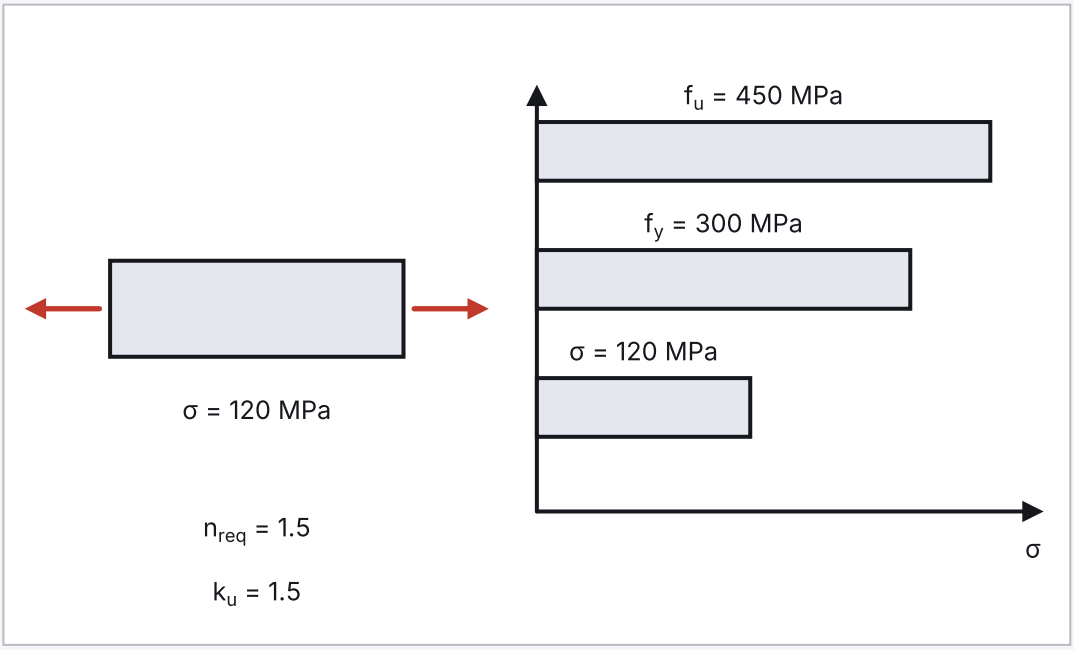


Figure 1 — A piece of the material under the applied stress, and the stresses as bars (drawn representatively, not compared). Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Factor of safety against yield and against ultimate	$n_y = f_y/\sigma$; $n_u = f_u/\sigma$
Governing factor, the ultimate margin scaled to the yield basis by $k_u \geq 1$	$n_{gov} = \min(n_y, n_u/k_u)$
Utilisation against the required factor	$u = n_{req}/n_{gov}$

Basis: Design factor and factor of safety[1](§1-10, 11th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
σ	Applied (equivalent) stress	120	MPa	spec: LOAD-SCHEDULE-01	0.01 … 5000 MPa
f_y	Yield strength	300	MPa	drawing: DRG-BR-701 rev A	1 … 5000 MPa
f_u	Ultimate tensile strength (optional)	450	MPa	drawing: DRG-BR-701 rev A	1 … 5000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_u	Ratio of the ultimate to the yield safety factor required	1.5		spec: SPEC-MECH-03 design criteria	1 … 3
n_{req}	Required factor of safety on yield	1.5		spec: SPEC-MECH-03 design criteria	1 … 10

3. Assumptions

A1. The load is static and the applied stress is the equivalent stress at the critical point. (Fatigue and stability are separate checks.)

4. Working

[5] $n_y = f_y / \sigma$ [1](§1-10)

= 300 MPa / 120 MPa

= 2.5

Factor of safety on yield

[7] $n_u = f_u / \sigma$

= 450 MPa / 120 MPa

= 3.75

Factor of safety on ultimate strength

[8] $n_{gov} = \min(n_y, n_u / k_u)$

= min(2.5, 3.75 / 1.5)

= 2.5

Governing factor (ultimate margin scaled to the yield basis)

[9] $u = n_{req} / n_{gov}$

= 1.5 / 2.5

= 0.6

Required over governing factor of safety

5. Results

SYMBOL	VALUE	DESCRIPTION
n_{gov}	2.5	Governing factor
n_y	2.5	Factor on yield
n_u	3.75	Factor on ultimate
u	0.60	Utilisation (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Required factor of safety against the governing factor	1.5	2.5	0.60	[1](§1-10)

7. Applicability

None.

8. Verification

Template mech.material-margin@1.1.0 Content hash 0b1c4346dd0d669e Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.